

Quadrotor Flyability, Mission & Compute Report

PX4 SITL + Gazebo Harmonic + ROS 2 Humble

Endurance | Factory navigation mission | Cloud vs edge compute

Version	v1.1.0
Date	2026-09-30
Repository	github.com/shaashwatmishra203-creator/Drone_sim
Airframe	n360_quad - quad X, 495 mm diagonal
Estimated AUW	3.117 kg (88% still unweighed)
Hover endurance	11.39 min measured in SITL
Factory mission	62 s, 109 m, 92% pack remaining
Compute	Edge required onboard; cloud for map sharing
Verdict	FLIES. Endurance is the constraint, not thrust.

What this report is, and is not

These results describe a **model** of the aircraft, validated in software-in-the-loop simulation. They do **not** establish that the real aircraft is safe to build, power or fly. No propeller has been spun, no pack discharged, and no thrust measured on a stand. Critically, **88% of the mass budget is still estimated rather than weighed**, and endurance moves roughly 2 minutes per 500 g. Treat every figure as a band, not a verdict.

Prepared with Claude Opus 5. All figures generated directly from config/airframe.yaml, out/sizing.json and the recorded flight logs.

1. Executive summary

The question was whether the parts list the team settled on can fly, and for how long. It can. Thrust-to-weight is about **3.45** and hovering consumes only **29%** of available thrust, which is comfortable margin. Flight capability is not the problem.

Endurance is. In still air the aircraft held a 5 m hover for **11.39 minutes** before reaching the 80% depth-of-discharge limit. In 6 m/s wind with gusts that falls to roughly 9 minutes. For an autonomy platform carrying two stereo cameras and a gimbal, that is a short working window.

Measured endurance	Hover power	Thrust-to-weight	Wind penalty
11.4 min	571 W	3.45	+22% power

Three findings block the build as specified

None of them is about flight dynamics:

#	Finding	Consequence
F1	The ZED 2 and ZED Mini cannot run on a Raspberry Pi 5. The ZED SDK requires an NVIDIA CUDA GPU; the Pi 5 has none.	Perception stack is non-functional as specified.
F2	The SpeedyBee F405 V5 cannot run a PX4 autonomy stack. PX4 dropped F405 support.	Nothing validated here transfers to that board.
F3	The 9x4.5 propellers reach ~15,900 rpm at full throttle against an estimated 16,111 rpm ceiling - about 1% margin.	13-inch props would add 57% endurance.

Evidence quality on F3

The 16,111 rpm ceiling is the **145000/D rule of thumb** for composite propellers, not a figure published by HQProp. This finding is only as strong as that number. Confirm the real rating with the manufacturer before treating it as settled - if their limit is higher, F3 shrinks from a blocking concern to a footnote.

2. The ROS 2 software

The simulation is driven by a small ROS 2 Humble package, `drone_eval`, living in `~/ws_px4/src/` alongside the standard `px4_msgs` and `px4_ros_com`. Data reaches ROS through PX4's uXRCE-DDS bridge rather than a `ros_gz` bridge, which is deliberately not installed - the PX4 path already carries everything needed and is lighter.

```
Gazebo Harmonic <---> PX4 SITL <---> MicroXRCEAgent <---> ROS 2
      (physics)      (flight control)      (DDS bridge)      drone_eval nodes
```

The two nodes

Node	What it does
flight_logger	Subscribes to position, attitude, status, per-motor commands and battery. Converts each motor command to rotor speed, then to thrust and shaft power through the propeller coefficients, and integrates a physical battery model. Writes one CSV per run and latches <code>/drone_eval/battery_empty</code> when the depth-of-discharge gate is reached.
mission_runner	Streams offboard heartbeats, arms, takes off, then flies the scenario profile - hover, forward-speed steps, maximum climb, or waypoints - and lands on completion, battery-empty, or timeout. PX4 requires a stream of setpoints <i>before</i> it will accept the offboard mode switch, so the node warms up before arming.

Why the logger computes power itself

PX4's SITL battery simulator drains the pack linearly on **wall clock** via `SIM_BAT_DRAIN` and never looks at the motors, so its `BatteryStatus` is useless for endurance. Instead the logger takes the real motor commands and integrates an open-circuit-voltage curve with internal-resistance sag. That matters because both effects shorten a flight and a naive energy-divided-by-power calculation misses them: as the pack empties its voltage falls, so the same power demands more current, which sags the voltage further.

Topics and QoS

Topic	Message	Direction
<code>/fmu/out/vehicle_local_position_v1</code>	<code>VehicleLocalPosition</code>	subscribe
<code>/fmu/out/vehicle_attitude</code>	<code>VehicleAttitude</code>	subscribe
<code>/fmu/out/vehicle_status_v4</code>	<code>VehicleStatus</code>	subscribe
<code>/fmu/out/actuator_motors</code>	<code>ActuatorMotors</code>	subscribe
<code>/fmu/out/battery_status_v1</code>	<code>BatteryStatus</code>	subscribe
<code>/fmu/in/offboard_control_mode</code>	<code>OffboardControlMode</code>	publish
<code>/fmu/in/trajectory_setpoint</code>	<code>TrajectorySetpoint</code>	publish
<code>/fmu/in/vehicle_command</code>	<code>VehicleCommand</code>	publish
<code>/drone_eval/battery_empty</code>	<code>std_msgs/Bool</code>	internal, latched

Two traps that cost real debugging time

QoS. PX4 offers every `/fmu/out/` topic as `BEST_EFFORT` with `TRANSIENT_LOCAL` durability. The rclpy default is `RELIABLE`, which is incompatible - a default subscriber receives nothing, silently. An empty log is a QoS mismatch until proven otherwise.

Topic names. PX4 1.18 appends message-version suffixes to some topics and not others, so both nodes resolve names against the live graph instead of hardcoding them. There is also no `/c lock` without a `ros_gz` bridge, so `use_sim_time` would freeze both nodes at zero; they use PX4 message timestamps, which are the simulation clock.

3. Parts and mass register

Every component, with the mass actually used by the model and a column for the real weight once the team measures it. **The measured column is deliberately empty: nothing has been weighed yet.** This is the single largest source of uncertainty in the entire report.

Component	Qty	Unit (g)	Est. (g)	Measured (g)	Confidence	Power (W)
Ovonic 22.2V 6S 50C 6000mAh (XT90 female)	1	830	830	—	estimated	-
HobbyWing XRotor 3115 900KV	4	195	780	—	estimated	-
Frame, PETG-CF printed, + nylon legs	1	550	550	—	estimated	-
3-DoF gimbal + gimbal camera	1	210	210	—	estimated	5.0
Stereolabs ZED 2	1	180	180	—	vendor spec	2.0
ALFA AWUS036ACM dual-band USB adapter + antennas	1	170	170	—	estimated	3.0
Wiring, XT90, standoffs, fasteners, straps	1	120	120	—	estimated	-
Raspberry Pi 5 (8 GB) + active cooler + 5V regulator	1	100	100	—	vendor spec	12.0
Stereolabs ZED Mini	1	62	62	—	vendor spec	2.0
HQProp 9x4.5 nylon composite	4	10	40	—	estimated	-
GPS / compass module + mast	1	40	40	—	estimated	0.5
SpeedyBee F405 V5 stack (FC + 4-in-1 ESC, 30x30)	1	35	35	—	vendor spec	2.0
TOTAL (all-up weight)			3117	—		26.5

Estimated versus real mass

2740 g of 3117 g (88% of all-up weight) is an engineering estimate, not a measurement. The four worth weighing first are the battery (830 g assumed), the four motors (195 g each assumed), the printed frame and legs (550 g assumed) and the gimbal assembly (210 g assumed). Endurance moves about 2 minutes per 500 g, so the true figure could plausibly sit anywhere between roughly 8 and 14 minutes. Filling in the measured column and re-running tools/sizing.py collapses that range to a single number.

Mass sensitivity

What the uncertainty above actually costs, for the current 9-inch build:

All-up weight (kg)	Hover power (W)	Endurance (min)	T/W
2.40	414	15.7	4.49
2.80	516	12.6	3.85
3.20	624	10.3	3.37
3.60	739	8.6	2.99
4.00	861	7.4	2.69

4. Propulsion, battery and mass properties

Motor	HobbyWing XRotor 3115 900KV x4 - 900 KV, 40 A continuous (all values unconfirmed)
Propeller	HQProp 9x4.5 nylon composite x4 - Ct 0.112, Cp 0.048, figure of merit 0.58
ESC	SpeedyBee F405 V5 4-in-1 ESC - 45 A continuous
Battery	Ovonic 22.2V 6S1P 50C 6000mAh - 6S, 6.0 Ah, 136 Wh at 22.7 V nominal
Discharge limit	80% depth of discharge (109 Wh usable)
Disc loading	19.0 kg per square metre (0.164 m2 total disc area)
Inertia (computed)	Ixx 0.0342, Iyy 0.0380, Izz 0.0582 kg m2
Centre of gravity	x +2.6 mm, y -1.6 mm, z +18.7 mm

Propeller comparison

Propeller	Disc load	Hover W	Endur. min	Gain	Max rpm	rpm margin	T/W
9x4.5 (current)	19.0	601	10.7	-	15927	1.1%	3.45
11x4.5	12.8	469	13.8	+29%	10310	21.8%	3.01
13x4.4	9.2	388	16.8	+57%	7334	34.2%	2.76

Disc load in kg per square metre. All configurations pass the thrust gates - the 9-inch rpm margin is the one marginal figure.

5. Flight test results

Two models were built deliberately independently - closed-form momentum theory, and PX4 flying the aircraft in Gazebo with power derived from real motor commands. They share only the configuration file. Their agreement is the reason the rest of these numbers are worth quoting.

Validation: analytical versus simulation

Quantity	Analytical model	Measured in SITL	Difference
Rotor speed (hover)	8569 rpm	8619 rpm	0.6%
Thrust per motor	779 g	788 g	1.2%
Hover power	601 W	571 W	5.0%
Endurance	10.7 min	11.39 min	6.3%

The residual gap is expected: the analytical side uses momentum theory with a figure of merit, while the logger integrates the power coefficient against actual rotor speed. They are different approximations, so exact agreement would be suspicious.

How long it flew

The aircraft held a 5 metre hover in still air until the battery model reached the 80% depth-of-discharge gate, consuming **107.5 Wh** of the 136 Wh pack:

Condition	Endurance	Power (W)	Current (A)	Rotor (rpm)	Thrust/motor (g)	Hold RMS (m)
Still air (default world)	11.39 min	571	25.5	8619	788	0.02
6 m/s wind + 3 m/s gusts	~9.2 min (proj.)	699	30.0	9237	907	0.93
Wind penalty	-19%	+22%	+18%	+7%	+15%	48x worse

Do not plan missions on the still-air number

The 11.4 minute figure is a benchmark measured with zero disturbance. In moderate wind the same aircraft loses about a fifth of its endurance and its position-hold error rises from 2 cm to nearly a metre. **Plan real missions around 7 to 8 minutes of usable time with reserve.**

Scenarios not yet run

This version covers still-air endurance and wind. Four scenarios are built and scripted but have not been executed: full 13-inch endurance, the Jetson compute variant, a forward-speed sweep, a maximum-climb test, and a forest waypoint mission. An interrupted 13-inch run did produce one useful reading - 384.8 W hover against the model's 388 W predicted, a 0.8% match that independently exercises the motor-constant derivation at a different propeller diameter. Re-run everything with `bash scripts/batch2.sh`.

6. Factory navigation and mapping mission

The representative mission, flown end to end: launch from base, slalom an obstacle course past storage racks and centreline pillars, fly a lawnmower scan pattern inside a 12 x 12 m room - the mapping pass whose data would go to the other vehicles - then return to base and land. Light wind (1.5 m/s, 0.8 m/s gusts) was applied after takeoff.

Airborne	Path flown	Energy used	Pack remaining
62.2 s	109 m	10.9 Wh	92.2 %

Leg breakdown

Leg	Time (s)	Share
Outbound (obstacle course)	18.5	32.2%
Scan (mapping pass)	17.5	30.5%
Inbound (return to base)	20.3	35.4%
Land	1.1	1.9%
TOTAL	57.4	

Scanning is a third of the flight

The mapping pass is **30% of the mission** - comparable to either transit leg. It is not a cheap add-on at the end, and mission planning should budget for it accordingly.

Operational margin

Metric	Value
Mean speed	1.76 m/s (peak 3.42)
Mean power	631 W (peak 1573)
Energy per metre flown	100 mWh/m
Usable energy consumed	10.0 %
Round trips per charge	10.0

Endurance is not the binding constraint for this task. The hover tests give about 11 minutes; this mission takes one. One pack supports roughly 10 round trips, which reframes the problem: for factory survey work the limit is mission planning and turnaround, not battery capacity.

What this mission does NOT establish

No obstacle avoidance was tested. The waypoints are pre-planned and collision-free by construction - the aircraft flew a known-good path, it did not perceive the racks and decide to avoid them. Likewise the scan leg measures what a mapping pass costs in time and energy; it does not produce a map. Both gaps need the depth pipeline, which needs CUDA hardware - see the next section.

7. Cloud versus edge compute

Where should perception and navigation run - on the aircraft, or on a server it streams to? The question is not which is faster in general, but three specific ones: **can the link carry it, can it stop in time, and what happens when the link drops?**

Verdict: split the workload

Edge and cloud are not competing options - they solve different halves of the problem. **Depth, visual-inertial odometry, obstacle avoidance and local planning must run onboard.** Map merging, global optimisation and distribution to the other vehicles belong off-board. A CUDA-capable board is required either way.

Bandwidth: what offload would have to push off the aircraft

Stream	Raw (Mbps)	Compressed (Mbps)
ZED 2 stereo (720p30, YUV422)	885	19.7
ZED Mini stereo (720p30, YUV422)	885	19.7
Depth map (720p30, float32)	885	110.6
TOTAL	2654	150

Depth compresses only about 8x - it is high-entropy, and lossy compression destroys exactly the geometry the obstacle avoidance depends on. That stream dominates the budget.

What the link actually delivers

Condition (ALFA AWUS036ACM, 802.11ac)	Goodput (Mbps)	Verdict
At base, line of sight (5 m)	390	OK
Mid obstacle course (16 m), steel racking	105	INSUFFICIENT
Room doorway (30 m), racking	60	INSUFFICIENT
Inside room (32 m), 1 wall + racking	11	INSUFFICIENT
Inside room, 2 walls (far corner)	2	INSUFFICIENT

The link is worst exactly where the mission matters most

The mapping pass happens **inside the room, behind a wall, at the furthest point from base**. Offload needs about 150 Mbps; there, the link delivers roughly 11 Mbps.

Latency: can it stop in time?

Architecture	Latency (ms)	React (m)	Stop (m)	Total (m)	Max safe (m/s)	Verdict
Edge (onboard)	69	0.12	0.27	0.39	3.39	SAFE
Cloud (regional DC)	578	1.02	0.27	1.29	1.71	UNSAFE
On-premise server	539	0.95	0.27	1.22	1.79	ACCEPTABLE

At the measured mission speed of 1.76 m/s, braking at 5.7 m/s². Corridor clearance is 1.25 m from centreline to the nearest rack or pillar. Note that moving the server on-premise saves only ~39 ms: the bottleneck is not network distance but that 150 Mbps does not fit down an 11 Mbps pipe, so bandwidth failure shows up as latency.

Availability: what a dropout costs

Link dropout	Distance flown blind
200 ms	0.35 m
500 ms	0.88 m
1000 ms	1.76 m

The strongest of the three arguments

An edge architecture is simply unaffected by a dropout: perception and control never leave the aircraft. In a steel-racked factory, multi-hundred-millisecond dropouts from shadowing and multipath are routine rather than exceptional. **This is a safety property rather than a performance one, and no amount of link engineering removes it.** A better radio raises the bandwidth ceiling and shaves latency; it does not make a dropout safe.

Why edge compute is what makes the map sharing work

The goal is for this drone to map a room and send that map to the other vehicles being designed. That requirement is the clearest argument for the split architecture:

	Size	Can it leave the aircraft?
Raw stereo + depth, continuous	2654 Mbps	No, at any range
Finished occupancy map of the room, per scan	~ a few hundred KB	Yes, over almost any link

Processing on the aircraft compresses the link requirement by roughly **four orders of magnitude**, turning an impossible stream into a trivial one. Edge compute is not an alternative to sharing the map - it is the precondition for it. The map is also the right thing to share: other vehicles need the result, not the sensor feed.

Power is not the deciding factor

Architecture	Total power (W)
Edge, Orin NX (map upload only)	26.5
Cloud offload, Pi 5 + full streaming	18.0
Cloud offload, Orin NX + streaming	31.0

The difference between the two realistic architectures is about 8.5 W - roughly 1.3% of the 630 W hover draw, or 15 seconds of flight time. Well inside the noise in the mass estimates.

Recommended architecture

ON THE AIRCRAFT (Jetson Orin NX) OFF-BOARD (server or cloud)

```
-----
ZED depth + visual-inertial odom -----> finished map (few hundred KB)
local occupancy map once per scan
obstacle avoidance + local planner <----- mission assignments,
flight control (PX4) global map, fleet coordination
```

must survive total link loss may be seconds late

Limits of this analysis

Verification level L0. No measurement on the real link, the real boards, or in the real building. Throughput figures are measured-order estimates for 802.11ac, not site survey data. A survey would change the numbers - access-point placement could raise the in-room figure substantially - but it would not change the dropout argument. Dropping to 480p15 would cut the offload bandwidth roughly 4x and make the numbers much closer, at the cost of obstacle-detection range and precision.

8. Recommended actions

Priority	Action	Cost	Benefit
1	Weigh the aircraft and fill in the measured column in section 3, then re-run the sizing tool.	free	Removes the largest uncertainty in every number here
2	Confirm HQProp's real maximum rpm for the 9x4.5.	free	Settles or dismisses finding F3 entirely
3	Move to 13-inch propellers. The 495 mm frame fits them with about 20 mm to spare - verify against CAD.	one prop set	+57% endurance, 10.7 to 16.8 minutes
4	Swap the Raspberry Pi 5 for a CUDA-capable board such as a Jetson Orin NX.	board	Makes the perception stack possible at all; costs ~0.6 min
5	Swap the SpeedyBee F405 for an H7 flight controller.	board	Makes a PX4 autonomy stack possible at all
6	Specify a dedicated 5 V / 5 A BEC and a power module.	small	Avoids brownout; enables real battery telemetry
7	Thrust-stand the motor and propeller to measure real Ct and Cp.	one day	Collapses the biggest remaining source of model error

What would change these conclusions

Measured masses materially different from the estimates, most likely the printed frame which could plausibly be 400 g or 800 g. An HQProp rating above roughly 17,000 rpm would retire F3. Measured propeller coefficients could move hover power by as much as 20%, carrying every endurance figure with them. None of these would threaten flyability itself - the thrust margin is large enough to absorb them.

A note on what the validation proves

The close agreement between the analytical model and the simulation demonstrates **internal consistency, not physical accuracy**. Both models read the same configuration file, so if the propeller coefficients are wrong then both are wrong together and would still agree with each other. Only a thrust stand can settle that. This is why item 7 above matters more than its low priority suggests.